

Project Proposal

Quantum Classifiers Against Classical Classifiers in Machine learning and Deep Learning

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1 Motivation

Quantum machine learning (QML) sits at the intersection of two of the most active areas in computing: near-term quantum algorithms and modern data-driven modeling. A central object in this field is the *variational quantum classifier* (VQC)—a parameterized quantum circuit trained as a supervised model, first proposed in the forms studied by Havlíček et al. [1], Schuld et al. [2], and Mitarai et al. [3].

The most important open empirical question in this area is also the most practically relevant one: under what conditions, if any, does a small trainable quantum model compete with standard classical classifiers? Recent large-scale benchmarks report a mixed picture—VQCs can match or exceed classical baselines on specifically constructed datasets [1], but often struggle to match even simple classical models of comparable parameter count on generic tasks. This tension is exactly where a well-designed experiment at our scale can make a useful contribution.

This project proposes a rigorous, experiment-driven comparison of a family of VQC architectures against carefully matched classical baselines on small classification problems. Rather than asserting a quantum advantage, the goal is to characterize the trade-offs among accuracy, circuit depth, parameter count, optimizer behavior, trainability, and noise sensitivity.

2 Central Research Questions

The project is organized around four concrete questions that can be answered with quantitative experiments at our scale:

1. **(Accuracy)** How does a VQC compare, on a like-for-like basis (matched trainable parameter count), to logistic regression, an SVM, a classical MLP, and to the *quantum kernel* / *QSVM* approach of Havlíček et al. [1] on small classification benchmarks?
2. **(Architecture)** How do the choice of feature map, entanglement topology, and circuit depth affect final accuracy and training stability?
3. **(Trainability)** Do we empirically reproduce the barren-plateau phenomenon of McClean et al. [6]—i.e., does the variance of the loss gradient decay as we scale the number of qubits and layers? Which ansätze mitigate it in our regime [7, 8]?
4. **(Noise)** How does performance degrade under realistic noise models? Is there any noise regime where VQCs remain stable while comparable classical models have no equivalent robustness concern?

3 Proposed Approach

3.1 Implementation stack

All experiments will use **Qiskit 1.x** with the community `qiskit-machine-learning` library . Classical baselines will be implemented in `scikit-learn` and PyTorch. Noisy simulation will use `qiskit-aer` with (i) synthetic depolarizing + readout noise models and (ii) calibrated **FakeBackend** snapshots of real IBM devices. All code, seeds, and hyperparameters will be tracked in a public GitHub repository for reproducibility.

3.2 Quantum circuit design

The VQC will follow the canonical encoding–ansatz–measurement structure:

Encoding (feature map). Three encoders will be compared:

- *Angle encoding:* $R_y(x_i)$ on qubit i . Simple baseline; classically simulable in closed form.
- *ZZFeatureMap* of Havlíček et al. [1]: Hadamards + diagonal Z and ZZ rotations parameterized by x_i and $x_i x_j$. Conjectured hard to simulate classically and the most widely used quantum-enhanced feature map.
- *Data re-uploading* : the input $\mathbf{x}$ is interleaved with trainable gates and re-uploaded multiple times, increasing effective expressivity without adding qubits.

Ansatz. Two families of hardware-efficient ansätze:

- **RealAmplitudes:** alternating R_y rotations and CNOT entanglers, real-valued amplitudes only.
- **EfficientSU2:** alternating R_y, R_z rotations with configurable entanglement topology (linear, circular, full).

Depth $L \in \{1, 2, 3, 4\}$ and entanglement topology will be swept. All parameters are initialized near zero following the Grant et al. [9] identity-block strategy.

Measurement and readout. Binary classification uses the expectation $\langle Z_0 \rangle$ on the first qubit, mapped through a sigmoid to a probability. Multi-class variants use softmax, implemented via Qiskit’s `SamplerQNN`.

Optimizer. COBYLA (gradient-free, small problem), SPSA (noise-robust stochastic)

3.3 Classical and quantum-kernel baselines

- **Logistic regression** (linear baseline, ℓ_2 -regularized).
- **SVM with RBF kernel** (strong nonlinear classical baseline).
- **MLP** with 1–2 hidden layers, matched to the VQC’s trainable parameter count to isolate the effect of the quantum structure rather than model capacity.
- **Quantum kernel + classical SVM** [1]: computes the Gram matrix using the same `ZZFeatureMap` but trains with a standard SVM. This isolates “quantum feature map” from “quantum trainable ansatz,” a distinction that is often blurred in the literature.

4 Datasets

Three small classification datasets at increasing difficulty:

1. **Two-moons**
2. **Iris (reduced)**
3. **Binary MNIST (0 vs 1) + PCA**

Each dataset will be split 70/15/15 (train/val/test) with fixed seeds.

References

- [1] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta. Supervised learning with quantum-enhanced feature spaces. *Nature*, 567(7747):209–212, 2019.
- [2] M. Schuld, A. Bocharov, K. M. Svore, and N. Wiebe. Circuit-centric quantum classifiers. *Physical Review A*, 101(3):032308, 2020.
- [3] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii. Quantum circuit learning. *Physical Review A*, 98(3):032309, 2018.
- [4] J. Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018.
- [5] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3(9):625–644, 2021.
- [6] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1):4812, 2018.
- [7] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles. Cost function dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications*, 12(1):1791, 2021.
- [8] M. Larocca, S. Thanasilp, S. Wang, K. Sharma, J. Biamonte, P. J. Coles, L. Cincio, J. R. McClean, Z. Holmes, and M. Cerezo. Barren plateaus in variational quantum computing. *Nature Reviews Physics*, 2025.
- [9] E. Grant, L. Wossnig, M. Ostaszewski, and M. Benedetti. An initialization strategy for addressing barren plateaus in parametrized quantum circuits. *Quantum*, 3:214, 2019.